EE 475 Introduction to Image Processing Homework 01

Rules:

- **Warning: NO LATE SUBMITTALS, NO E-MAIL SUBMISSIONS.** Please check out the deadline and arrange yourself accordingly (and welcome to the real life).
- If asked, any code must be supported with concise comments for evaluation. Please remember that too many or long comments make the code intractable whereas a code without or too little comments makes it incomprehensible.
- Please submit the pdf file of your answers to Moodle web page.
- **Specifically starting from this homework,** I would like to see appropriate embedding of graphics in your report. Whether you are using LaTeX or MS–Word please learn how to appropriately place the figures in your document. While this is straightforward for LaTeX, my suggestion for MS–Word would be to place the figures within a text box that also contains the figure caption. This makes moving them around easier.

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1. **Scaling:** Given an image as a matrix A,
   (a) Provide theoretically a function that will create a new image scaled to the range to $[0,\ 1]$. Which functions are needed for this normalization process?
   (b) Write a code in your preferred programming language that will implement the procedure.
2. **Interpolation:** The provided image files of the chronometer are obtained from the original file `Chronometer.tif` by sub–sampling the original file at each direction.
   (a) What are the resolutions of the provided image files?
   (b) Implement a linear and a bi–cubic interpolation routine to obtain from the reduced resolution images the original image.
   (c) Devise a performance measure for the interpolation algorithms. 'Judge' the performance of your algorithms and provide a 100 words report with table(s) and figure(s) if necessary. The report should contain the code(s) your created in an appendix.